Assessing changes in algal population size and composition in the UC Davis Arboretum waterway

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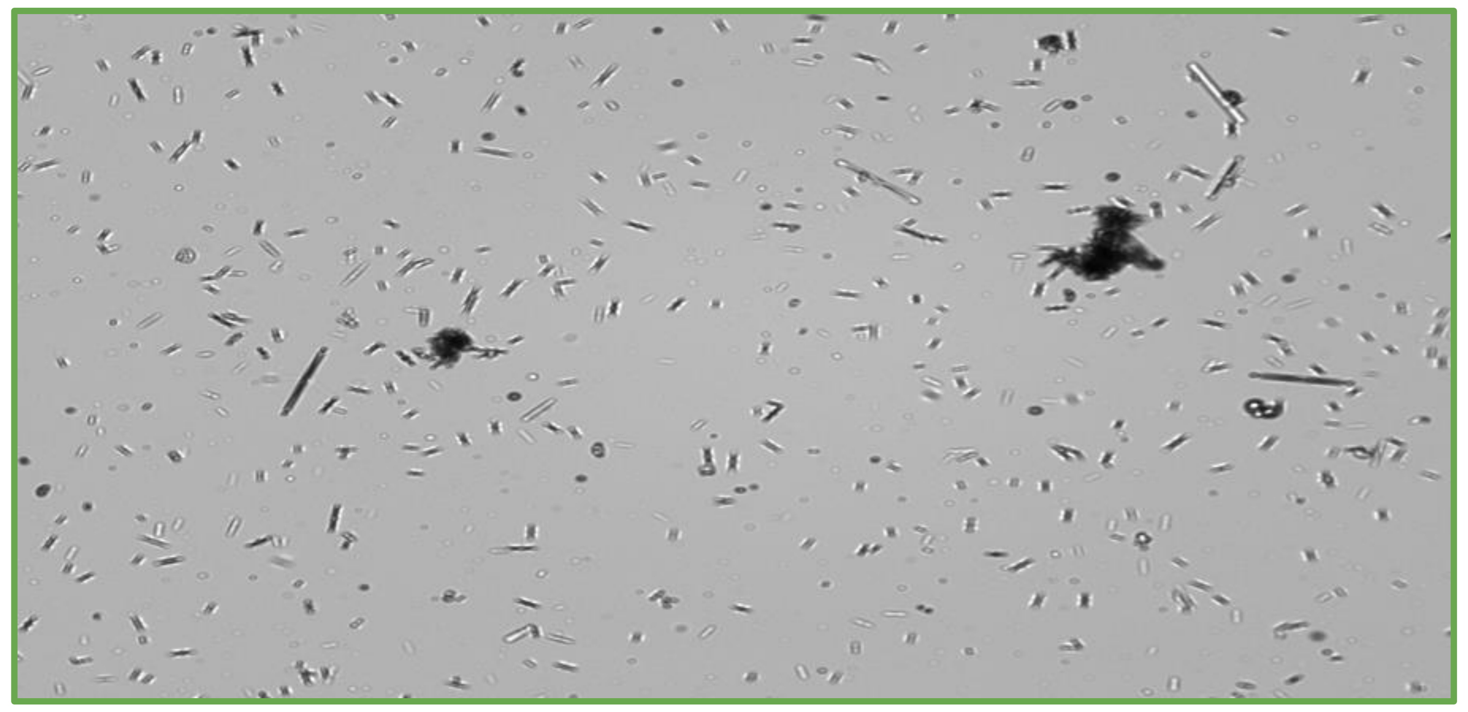
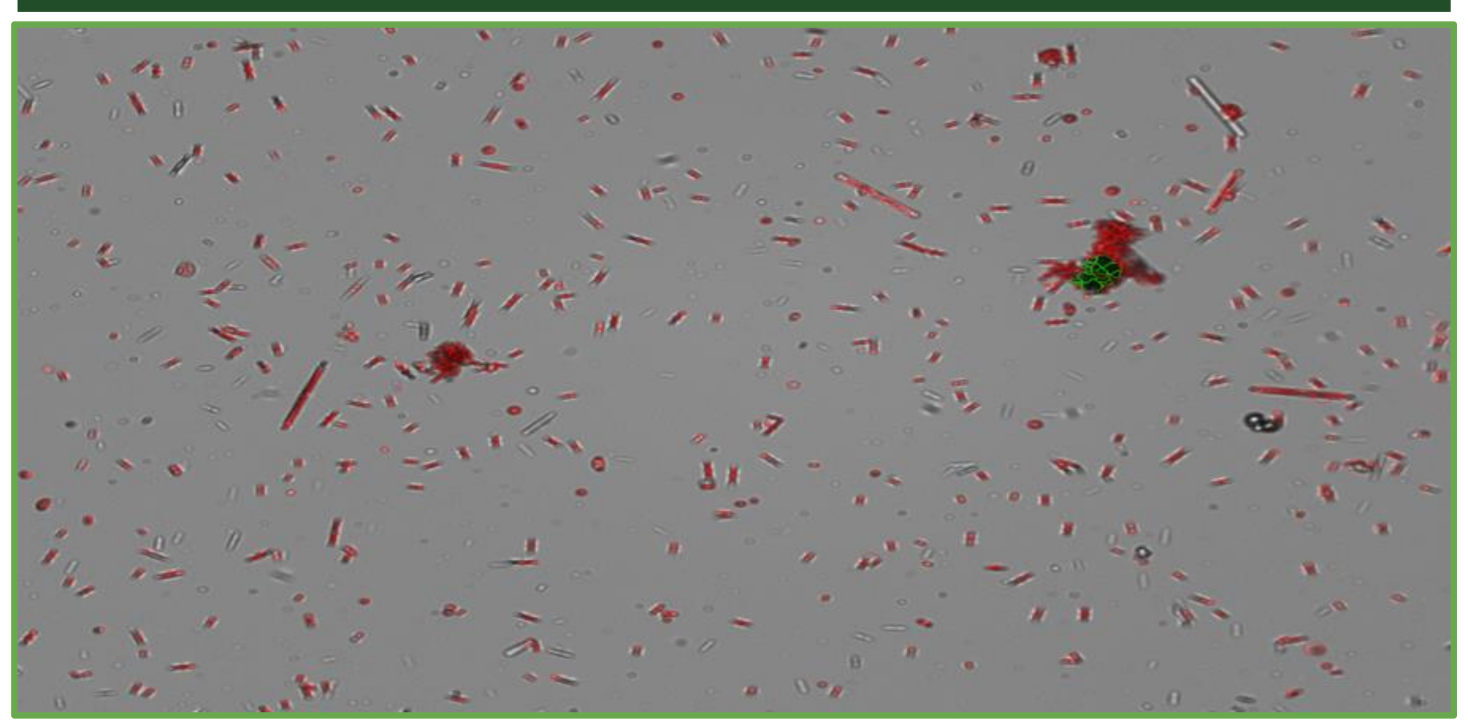
Introduction

Uncontrollable algal blooms-sometimes toxic (HABs) have been becoming more and more frequent all around the U.S, including northern California [1] [3]. They have been responsible for events such as oxygen depletion, producing toxins in the water, and outcompeting other crucial species in the environment [1]. Some freshwater algae species most common to the area of research and the ones (most likely) found were varieties of diatoms and dinoflagellates and green algae (Chlorophyta) [1] [2]. It is also possible that cyanobacteria (not classified as algae) was found since it is common to the area as well. In order gain a better understanding on algae species and other photosynthetic organisms in the urban northern California environment, the population densities at locations along the UC Davis arboretum were measured to pinpoint what may cause algae populations to fluctuate. It was hypothesized that the progression of colder winter weather to warmer, spring weather would raise population levels. After conducting the experiment, it seems that there are numerous and other significant influences that contribute to algal populations then previously thought. By learning what effects algae populations, conservationists and other groups can implement ways of controlling algae in ways that's safer for an ecosystem in similar urban environments.



This is the view the DeNovix cell drop machine gives. It allows the viewers to see organisms moving in real time. This is sample from week 1 site 3



The cells that contained photosynthetic pigments are highlighted in red by the cell drop machine allowing it to approximate the number of algal cells.

Possible influences on population size:

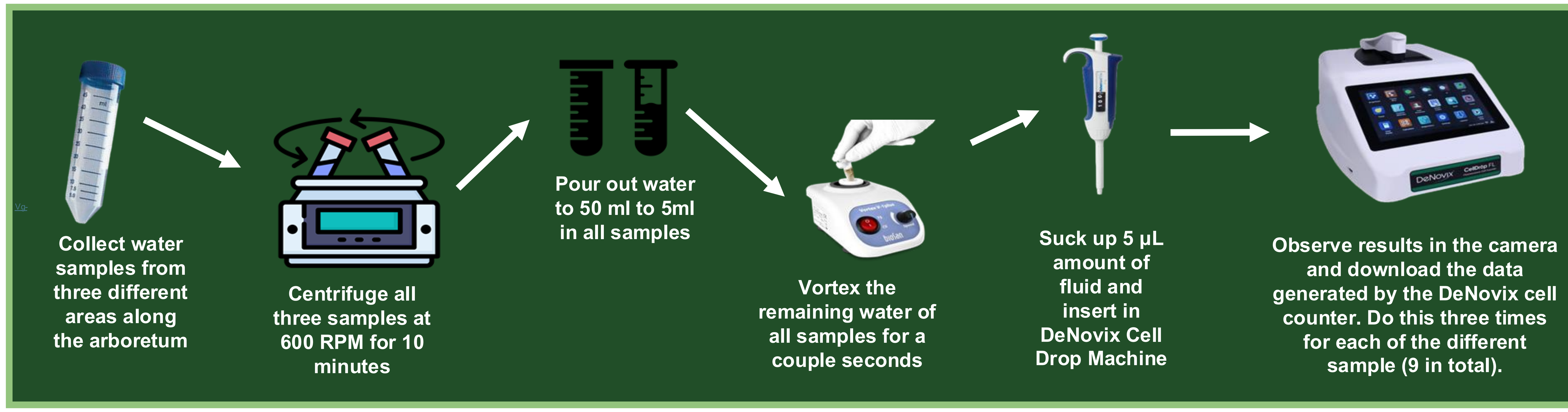
Week 1: It was 59 °F and partly cloudy during collection. Note that collection site 3 (green tube) had a foam blocker that may have blocked the flow of particles on the water's surface

Week 2: It was 57°F and was raining during collection. The foam blocker at collection site 3 was not present.

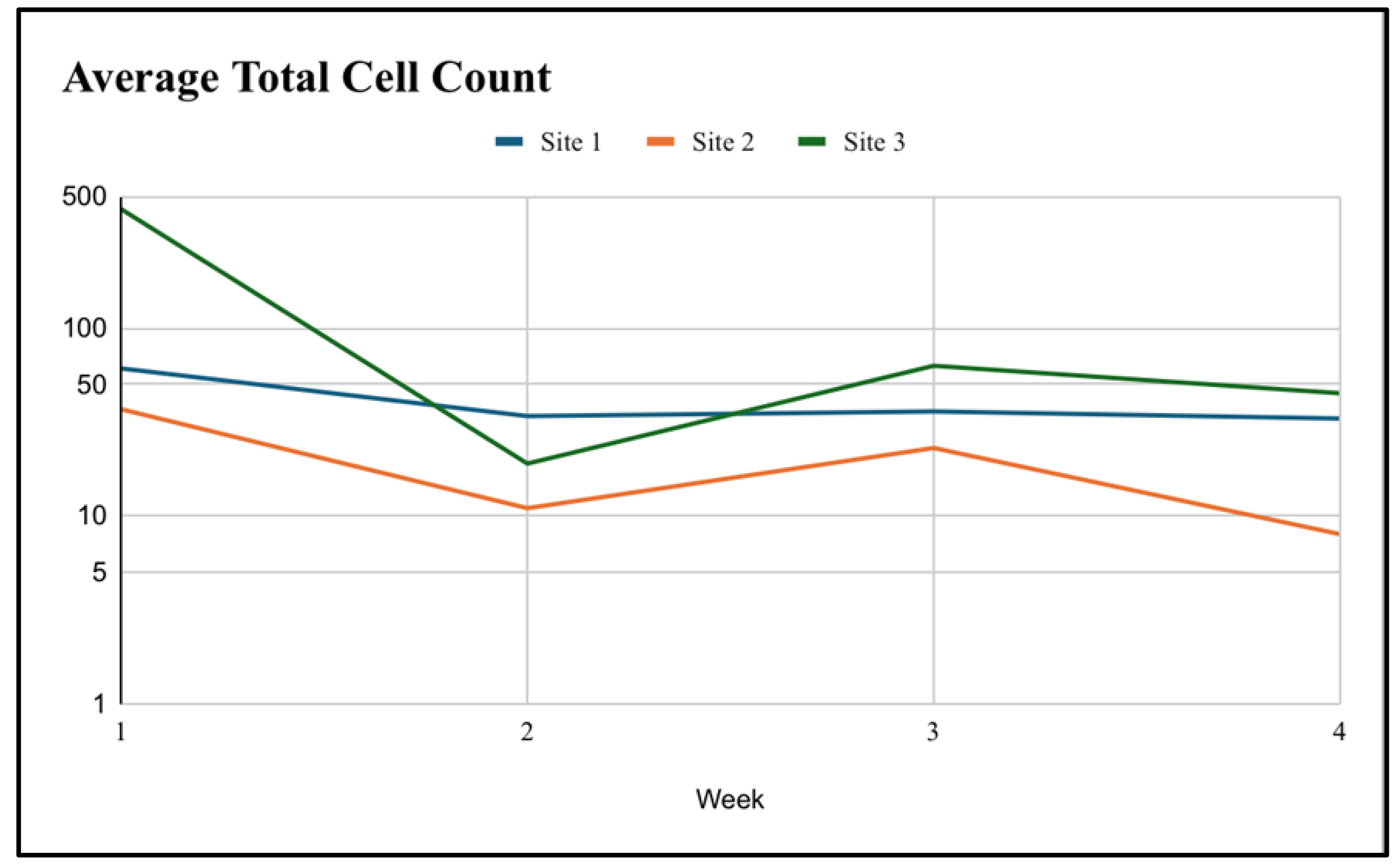
Week 3: It was 61°F and overcast during collection.

Week 4: It was 65°F during collection and partly cloudy.

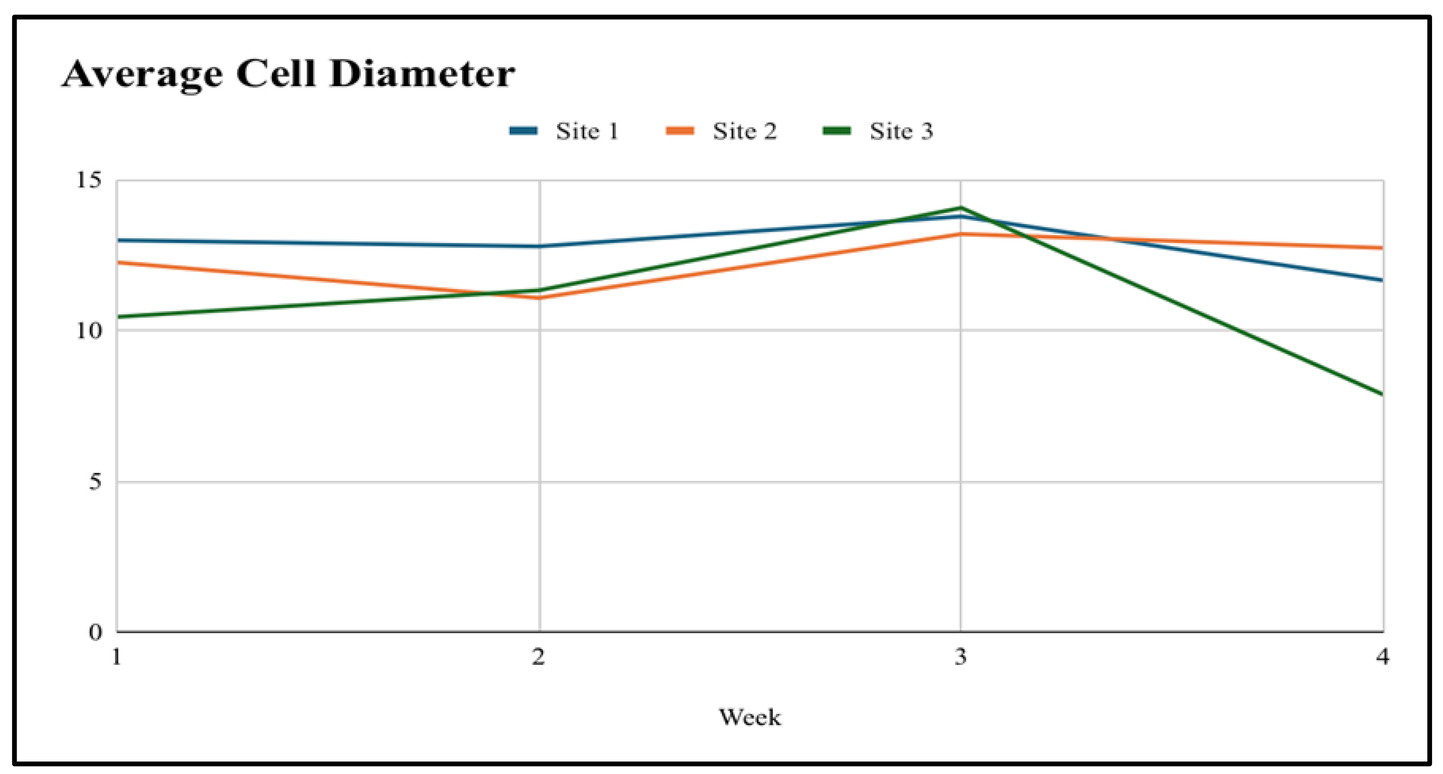
- Over between March 1st and April 2nd, the average temperature rose from 64°F to 70°F
- Between March 1st and April 2nd, the amount of sunlight outside gradually increased 1 hour and 8 minutes.



This diagram shows how to project was run. In all, 3 50ml Falcon tubes were needed to collect water , a centrifuge and vortex machine used to homogenize the water, an adjustable pipette and pipette tips to squirt the water, and the DeNovix cell drop counter to visualize and count the findings.



This graph represents the averages taken From March 5 to April 2



This graph represents the average cell diameter from March 5 to April 2

Results & Discussion

After observing the data, it was found that the changes in density of photosynthesizing organisms at the places sampled were most likely due to water circulation, sunlight availability, and the amount it rained during the time period. To start, Site 3 was visually the most stagnant out of all 3 sites and was also the most densely populated. Additionally, it was observed that the site had a foam top on the water's surface, which looked like it caused a buildup of nutrients and other debris on the water's surface and slowed the flow of water on the surface. The combination of the two factors are most likely what contributed to the cell count being so high the first collection at site 3 compared to the others. Second, is sunlight. During the weeks of collection, the temperatures continued to rise but unlike what was previously hypothesized to happen, the population didn't have remarkable changes in growth. One factor that wasn't considered was the weather. Weather data from March 7 through April 2 showed that nearly half of the days throughout the experiment, it was cloudy. Since the DeNovix machine solely calculated organisms who had photosynthetic pigments, the amount of sunlight was no doubt crucial to the density of organisms accounted for. Lastly, it is believed that rain influenced our findings. The reason being that the waterway collects the excess water throughout the campus when it rains and during the water's journey, the it is able to pick up nutrients capable of stimulating growth such as nitrogen and phosphorous. Between weeks 2 and 3, it rained twice, around .4 inches both instances. 2 of the 3 sites saw increases in their densities during this period which is possibly due to said weather conditions.

All in all, ecosystem dynamics and water chemistry are incredibly complex and difficult to understand. Using the data and knowledge collected, it is believed that the changes in the average cell count of photosynthesizing organisms at our sites along the arboretum waterway were likely due to differences in water movement, sunlight, and weather. This information can be useful to conservationists and other groups trying to balance ecosystems and prevent toxic and excessive algae growth in waterways.

References

[1] Wehr, John D, et al. *Freshwater Algae of North America : Ecology and Classification*. Amsterdam ; Boston, Elsevier/Ap, Academic Press Is An Imprint Of Elsevier, 2015.
[2] Algal Blooms - Water Education Foundation." *Water Education Foundation*, Water Education Foundation, 22 June 2020
[3] "Freshwater Flows, Reduced Diversions Help Abate Toxic Algae." NRDC, 23 Sept. 2019, www.nrdc.org/bio/melanie-sturm/freshwater-flows-reduced-diversions-help-abate-toxic-algae.

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